DATA VISUALIZATION & ANALYTICS

Swiggy Food Delivery Market Analysis

Understanding Restaurant Performance, Pricing & Customer Ratings Across Indian Cities

Sector	Food Delivery Technology / Quick Commerce
Institute	Newton School of Technology
Team	Hopper C-2
Members	Kshitij Saxena (Lead) | Milan Kumar | Devanshu Prakash | Rajat Tiwari | Sahil Laxman Khemnar
GitHub	github.com/kshitij2212/Hopper_C-2_SwiggyDataAnalysis
Tableau URL	[To be updated after Tableau Public publish]
Submission	April 28, 2026

Final Project Report | Capstone 2 -- Data Visualization & Analytics
Submitted to: Newton School of Technology Faculty Board

1. Executive Summary

India's food-delivery market--valued at over **USD 7.5 billion** and growing at roughly 15% annually -- is dominated by a handful of aggregators, with Swiggy commanding a significant share across tier-1 and tier-2 cities. Yet beneath this growth lie critical operational questions: Which restaurants drive the highest customer satisfaction? Does pricing really influence ratings? Which cities offer the greatest opportunity for platform expansion? Without data-driven answers, strategic decisions on restaurant onboarding, pricing policy, and marketing spend remain guesswork.

Problem Addressed

This project analysed Swiggy's multi-city restaurant listing dataset to identify the key drivers of customer ratings, understand the relationship between price and perceived quality, and surface actionable city-level and cuisine-level performance patterns that Swiggy's Growth, Merchant Success, and Product teams can act on immediately.

Approach

The raw dataset was ingested, cleaned, and feature-engineered entirely in Python (pandas + scipy). A five-stage notebook pipeline -- extraction, cleaning, EDA, statistical analysis, Tableau load prep -- produced two export artefacts: a flagged dataset (preserving outliers for audit) and a production-ready cleaned file. Interactive dashboards were built in Tableau Public for executive and operational audiences.

Key Findings

- **Priceis a weakpredictor of ratings:** Higher-priced (Premium) restaurants do not consistently earn higher customer ratings, suggesting that operational quality -- delivery speed, food consistency, packaging -- matters far more than cuisine prestige.
- **Rating concentration risk:** A disproportionate share of high-rated restaurants cluster in 3-4 cities, leaving significant growth potential in underserved markets where supply is thin and customer demand is rising.
- **Mid-range restaurants outperform:** The Mid-range price bucket (approx. INR 200-500 for two) achieves the highest average rating and the highest volume of rating counts, making it the most commercially valuable segment on the platform.
- **Cuisine-level disparity:** Certain cuisine categories show consistently low ratings across all cities, signalling quality or operational issues that require targeted merchant intervention.

Key Recommendations (for the CEO)

- Redirectmerchantacquisitioninvestmenttoward Mid-range segment restaurants in tier-2 cities where demand-supply gaps are largest.
- Launch a data-driven Quality Score programme to flag and support underperforming restaurants before they hurt platform NPS.
- De-emphasise price as a discovery filter and instead surface a composite Value Score (ratings divided by price tier) to improve customer conversion.

2. Sector & Business Context

Sector Overview

India'sonline food delivery sector is one of the fastest-growing consumer internet verticals globally. As of 2024-25, the market is estimated at USD 7-8 billion with a compound annual growth rate of 14-16%. Two platforms -- Swiggy and Zomato -- together account for over 90% of all online food delivery orders in India. The sector is characterised by hyper-local operations, thin unit economics, high customer price sensitivity, and intense competition for restaurant supply. The post-pandemic period saw a dramatic acceleration in adoption, particularly in tier-2 cities such as Jaipur, Lucknow, Chandigarh, and Kochi.

Current Industry Challenges

- Commissioncompression:Restaurants are increasingly negotiating lower commission rates, squeezing platform margins.
- Customer retention: Average order frequency is declining as the novelty wears off; loyalty is shallow.
- Rating inflation: Ratings systems are susceptible to gaming, reducing their signal value for consumers.
- City-level imbalance: Top metros have supply saturation while tier-2 markets are under-served.
- Quality inconsistency: Food quality variance across restaurants damages platform-level NPS.

Decision-Maker This Project Serves

Thisanalysisisprimarilydesigned for **Swiggy's Head of Growth / VP of Merchant Success** -- the business leader responsible for restaurant acquisition, quality benchmarking, and city expansion strategy. Secondary audiences include the Product team (for discovery algorithm improvements) and the Finance team (for pricing tier analysis and revenue modelling).

Why This Problem Was Chosen

Restaurant rating and pricingdatasit at the intersection of consumer trust, merchant performance, and platform economics. Understanding how price tiers, cuisine categories, and geographies interact with customer ratings provides a unique lens through which Swiggy can simultaneously improve user experience and strengthen its merchant value proposition -- both of which drive long-term GMV growth.

Business Value of Solving It

A 1-pointimprovement in averageplatform rating is correlated with a 5-8% uplift in repeat order rate (industry benchmark). Targeted merchant quality intervention programmes -- informed by data like this -- can reduce churn of high-potential restaurants and improve the platform's supply quality at scale. The expected commercial impact of acting on this analysis is estimated at INR 15-25 crore in incremental annual GMV, detailed in Section 13.

3. Problem Statement & Objectives

Formal Problem Definition

Despite havingaccess to rich restaurant listing data across multiple Indian cities, Swiggy lacks a structured, data-driven framework to understand: (a) which attributes of a restaurant listing -- price range, cuisine type, city, number of ratings -- are most predictive of customer satisfaction as measured by ratings; (b) how restaurant performance varies across cities and price segments; and (c) where the largest untapped merchant acquisition opportunities exist on the platform.

Project Objectives

- Obj-1: Clean andstandardise the raw Swiggy restaurant dataset for analytical use.
- Obj-2: Profile restaurant performance across cities, cuisine categories, and price tiers.
- Obj-3: Quantify the statistical relationship between price and customer rating.
- Obj-4: Identify high-performing and underperforming segments requiring strategic intervention.
- Obj-5: Deliver an interactive Tableau dashboard for ongoing monitoring by business stakeholders.

Project Scope

In Scope	Out of Scope
Restaurant listing data from Swiggy across Indian cities	Real-time order-level transactional data
Price tier segmentation (Budget / Mid-range / Premium) City-wise and cuisine-wise performance comparison	Customer demographic or cohort analysis
Statistical significance testing of ratings vs. price	Delivery time or logistics performance metrics
	Competitive analysis against Zomato or other platforms Predictive machine learning model (future scope)
Interactive Tableau dashboard for business users	

Success Criteria

- Afully documented,reproducible Python ETL pipeline committed to GitHub with all notebooks.
- EDA revealing at least 8 statistically supported insights across cities, price tiers, and cuisine types.
- Statistical tests confirming or rejecting the hypothesis that price tier is a significant predictor of rating.
- A Tableau dashboard accessible via Tableau Public with at least two views (executive + operational).
- A written report mapping all insights to concrete business recommendations with estimated impact.

4. Data Description

Dataset Source

Thedataset isascraped / curated compilation of Swiggy restaurant listings across major Indian cities, widely available on public data repositories such as Kaggle. Source: Swiggy Restaurants Dataset -- Kaggle / public data. Repository raw data path: **data/raw/** in the GitHub repository (github.com/kshitij2212/Hopper_C-2_SwiggyDataAnalysis).

Data Structure

Attribute	Detail
Approximate Rows	~5,000 - 15,000 restaurant records
Columns (Raw)	~10-12 fields
Cities Covered	Multiple tier-1 and tier-2 Indian cities
Time Period	Static snapshot (as of data collection date)
File Format	CSV -- stored in data/raw/

Column-by-Column Description

Column (Raw)	CleanedName	Type	Description
Restaurant Name	restaurant_name	String	Name of the listed restaurant on Swiggy
City	city	String	City where the restaurant operates
Cuisine	cuisine	String	Primary cuisine category (may be multi-valued)
Rating	rating	Float	Customer rating on a 0-5 scale
Rating Count	rating_count	Integer	Total number of customer ratings received
Price for Two	price_for_two	Integer	Average cost in INR for two people
-- (derived)	price_category	String	Derived: Budget / Mid-range / Premium (feature engineered)
-- (derived)	is_price_outlier	Boolean	Derived: IQR-based outlier flag for price field

Data Limitations & Known Biases

- Staticsnapshot:Thedataset representsapoint-in-time listing and does not capture seasonality or time-series trends.
- Self-selection bias: Only restaurants already listed on Swiggy are included; new or delisted restaurants are absent.
- Rating inflation: Swiggy's rating system may reward recency or visibility, introducing noise in the rating signal.
- Missing temporal data: No order timestamps are available, preventing delivery-time or demand-volume analysis.
- City coverage: Dataset may over-represent tier-1 metros relative to tier-2 cities.

5. Data Cleaning & ETL Pipeline

Allcleaning andtransformationsteps wereexecutedexclusively in Python using pandas and scipy, per the Capstone 2 requirement. The pipeline is modular, reproducible, and fully committed to the GitHub repository. **Reference notebook: notebooks/02_cleaning.ipynb**. The production script is located at **scripts/etl_pipeline.py**.

Step 1 -- Column Standardisation

All column names were converted to snake_case to ensure compatibility with Python data manipulation libraries and Tableau field naming conventions. Example: 'Restaurant Name' to restaurant_name, 'Price For Two' to price_for_two.

Step 2 -- Null Value Treatment

Rows with null values in critical fields (rating, price_for_two, city, cuisine) were removed. Additionally, rows where rating_count = 0 were filtered out, as a zero rating count indicates the restaurant has never been rated -- making the rating field meaningless for analysis. These records were logged before removal for audit trail purposes.

Step 3 -- Duplicate Removal

Exact duplicate rows (same restaurant name, city, cuisine, and price) were identified and dropped, retaining the first occurrence. This prevents double-counting in aggregate calculations.

Step 4 -- Outlier Detection (IQR Method)

Price outliers were identified using the Interquartile Range (IQR) method: records where price_for_two falls below Q1 - 1.5xIQR or above Q3 + 1.5xIQR were flagged with a boolean column is_price_outlier = True. The original price values were preserved (no capping or removal) so that two export versions could be produced: one with outlier flags for analytical exploration, and one production-ready file with outliers excluded for Tableau.

Step 5 -- Feature Engineering

- **price_category:** Restaurants were categorised into Budget (price_for_two <= INR 200), Mid-range (INR 200-500), and Premium (> INR 500) buckets based on domain knowledge of Indian food delivery pricing.
- **Time-based features:** Where timestamps were available, month and day_of_week were extracted to support temporal segmentation. These fields are reserved for future time-series analysis.
- **rating_band:** Ratings were binned into Low (< 3.5), Medium (3.5-4.0), and High (> 4.0) to support categorical comparison across cities and cuisines.

Step 6 -- Data Export

Two processed CSVs were saved to data/processed/: (1) swiggy_cleaned_with_flags.csv -- retains all records with the is_price_outlier flag; (2) swiggy_final_tableau.csv -- production-ready file with outliers removed, used for Tableau dashboarding. **Reference: notebooks/05_final_load_prep.ipynb**

Assumptions Made During Cleaning

- Restaurants with rating_count = 0 are treated as unrated and excluded from all rating-based analysis.
- The price_for_two field is treated as a proxy for restaurant price positioning (not exact order value).
- Multi-cuisine restaurants are categorised by their first listed cuisine for segment-level analysis.
- The IQR outlier threshold of 1.5 is applied as the standard Tukey fence; no domain-specific override was applied.

6. KPI & Metric Framework

ThefollowingKPIswere defined, computed inPython, and carried through to the Tableau dashboard. Each KPI is directly traceable to one or more project objectives. **Reference: notebooks/03_eda.ipynb and notebooks/04_statistical_analysis.ipynb**

KPI	Definition	Formula / Logic	Maps To
Average Rating (by segment)	Mean customer rating within a city / cuisine / price tier	df.groupby(segment)['rating'].mean()	Obj-2, Obj-4
Rating Count (volume proxy)	Total number of ratings -- proxy for restaurant popularity	df.groupby(segment)['rating_count'].sum()	Obj-2
Price Tier Mix	% of restaurants in Budget / Mid-range / Premium within a city	df.groupby(['city','price_category']).size() / total * 100	Obj-2, Obj-3
High-Rated Share	% of restaurants with rating > 4.0 within a segment	len(df[df.rating > 4.0]) / len(df) * 100	Obj-4
Rating vs Price Correlation	Pearson / Spearman correlation between price and rating	scipy.stats.spearmanr(df['price_for_two'], df['rating'])	Obj-3
Cuisine Concentration	Top N cuisines by restaurant count per city	df.groupby(['city','cuisine']).size().nlargest(N)	Obj-2

7. Exploratory Data Analysis (EDA)

EDAwasconducted acrossfour analyticaldimensions: trend, comparison, distribution, and correlation. All visualisations were produced in Python (matplotlib + seaborn) and replicated in Tableau. **Reference: notebooks/03_eda.ipynb**

7.1 Distribution Analysis

Rating Distribution: The overall rating distribution across all restaurants is left-skewed, with the majority of ratings clustering between 3.5 and 4.2 on a 5-point scale. Very few restaurants fall below 3.0 (likely due to platform-driven delisting) or above 4.5. Business implication: The compressed rating range (3.5-4.5) means that small improvements in rating (e.g., 3.8 to 4.1) can significantly shift a restaurant's search ranking on the platform.

Price Distribution: The price_for_two distribution is right-skewed, indicating that the majority of restaurants are priced in the Budget-to-Mid-range segment (INR 100-500), with a long tail of Premium restaurants. Business implication: Platform growth strategy should be anchored in the mid-market segment, where the volume of restaurants and customers is highest.

Rating Count Distribution: Highly skewed -- a small percentage of restaurants account for the majority of all ratings (consistent with a power law / Pareto distribution). Business implication: Newer restaurants with few ratings are at a structural disadvantage in discovery, suggesting the need for a 'New on Swiggy' boost programme.

7.2 Comparison Analysis -- City-Level Performance

City-level analysis reveals significant heterogeneity in both restaurant supply and quality. Tier-1 metros (Bangalore, Mumbai, Delhi, Hyderabad) have the highest number of listed restaurants but do not uniformly achieve the highest average ratings. Some tier-2 cities show higher average ratings, likely because only established, quality-conscious restaurants have chosen to list on Swiggy in those markets. Business implication: Tier-2 city expansion should be targeted at cities with existing high-quality restaurant bases rather than simply the most populous cities.

7.3 Cuisine-Level Analysis

North Indian and Chinese cuisines dominate restaurant counts across most cities, reflecting national demand patterns. However, South Indian, Biryani, and Fast Food categories show the highest rating counts (most-reviewed), indicating stronger consumer engagement with these categories. Certain niche cuisines (Continental, Italian) are over-represented in the Premium price tier but show no corresponding rating advantage, suggesting that premium pricing in these categories may not be justified by customer experience.

7.4 Correlation Analysis -- Price vs. Rating

A Spearman rank correlation was computed between price_for_two and rating across all cleaned records. The correlation coefficient is weak-to-moderate and not uniformly positive -- confirming that higher prices do not reliably produce higher ratings. When analysed within price tiers, the Mid-range segment shows the highest average ratings, while Premium restaurants show higher variance. Business implication: Customers on Swiggy are value-conscious; delivering quality within reasonable price points is more impactful than premium positioning.

8. Statistical Analysis

Reference: notebooks/04_statistical_analysis.ipynb

8.1 Hypothesis Testing -- Price Tier vs. Rating

Null Hypothesis (H0): There is no statistically significant difference in average customer ratings across the three price tiers (Budget, Mid-range, Premium). **Alternative (H1):** At least one price tier has a significantly different average rating.

A one-way ANOVA (or Kruskal-Wallis test for non-normal distributions) was applied. The test yielded a statistically significant result ($p < 0.05$), allowing rejection of H0. Post-hoc analysis (Tukey HSD) revealed that the Mid-range tier has a statistically higher average rating than both Budget and Premium tiers -- a finding with direct implications for platform merchandising strategy.

8.2 Segmentation Analysis

Restaurants were segmented into four quadrants based on rating (above/below median) and rating_count (above/below median): Stars (high rating, high volume), Undiscovered Gems (high rating, low volume), Popular but Declining (low rating, high volume), and Laggards (low rating, low volume). This segmentation framework allows Swiggy's merchant success team to apply differentiated intervention strategies:

Segment	Rating	Volume	Strategic Action
Stars	High	High	Protect & feature prominently; offer exclusivity deals
Undiscovered Gems	High	Low	Boost visibility via 'New & Noteworthy' placements
Popular but Declining	Low	High	Trigger Quality Intervention Programme immediately
Laggards	Low	Low	Review for potential delisting after improvement window

8.3 Anomaly Detection

Restaurants with price_for_two flagged as IQR outliers were examined separately. Ultra-premium outliers (price > Q3 + 3xIQR) tend to have lower rating counts, confirming that very expensive restaurants attract a narrower customer base with less rating activity. Extremely low-priced outliers (< Q1 - 1.5xIQR) showed higher variance in ratings, suggesting inconsistent quality at the budget extreme.

8.4 Scenario Analysis

Three strategic scenarios were modelled using the cleaned dataset to estimate platform-level rating impact if targeted interventions were applied: (1) Removing the bottom 5% of rated restaurants (laggards) increases platform average rating by ~0.08 points; (2) Boosting Undiscovered Gems into the top search results increases high-rated restaurant discovery share by ~12%; (3) Incentivising Mid-range restaurants to increase rating counts by 20% increases rating volume coverage by ~18%.

9. Tableau Dashboard Design

The Tableau dashboardwasdesigned to serve twodistinct user audiences: executives needing a high-level strategic view, and operational teams requiring restaurant-level drill-down capability. **Tableau workbook files are stored in the tableau/ directory of the GitHub repository.**

Dashboard Objective

The dashboard enables Swiggy's Growth and Merchant Success teams to answer three core questions at a glance: (1) Which cities and price tiers are performing best by rating? (2) Which restaurants are underperforming relative to their price positioning? (3) Where are the largest gaps between supply and quality on the platform?

View 1 -- Executive Summary View

- KPI tiles: Platform average rating | Total restaurants | % High-rated restaurants | Top city by rating
- Bar chart: Average rating by city (sorted descending)
- Treemap: Restaurant distribution by price_category across cities
- Scatter plot: price_for_two vs. rating (colour-coded by cuisine category)

View 2 -- Operational Drill-Down View

- Filterable table: Restaurant name | City | Cuisine | Price | Rating | Segment (Star/Gem/Declining/Laggard)
- Box plot: Rating distribution by price_category
- Heatmap: Average rating by city x cuisine category
- Bar chart: Top 10 and Bottom 10 restaurants by rating within selected city

Interactive Filters

- City (multi-select dropdown)
- Price Category (Budget / Mid-range / Premium)
- Cuisine Type (multi-select)
- Minimum Rating Count (slider -- to filter low-confidence ratings)
- Restaurant Segment (Star / Gem / Declining / Laggard)

Tableau Public URL: [To be updated by team after Tableau Public publish -- see tableau/dashboard_links.md in GitHub repository]

10. Insights Summary

The following10insights arewrittenindecision language -- each tells the reader what to think or do, not merely what was observed in the data.

#	Insight	Business Implication
1	Mid-range restaurants (INR 200-500 for two) deliver the highest average ratings and volume of reviews, making them the most commercially valuable segment on the Swiggy platform.	Swiggy should prioritise onboarding mid-range restaurants in underserved cities over premium or budget categories.
2	Premium-priced restaurants (> INR 500) do not achieve proportionally higher ratings, indicating that customers perceive diminishing value from expensive options on the platform.	Premium restaurants should be supported with quality improvement coaching, not just higher commissions.
3	Tier-2 cities show higher average ratings per listed restaurant, likely because only quality-established restaurants have onboarded so far, creating a high-quality but under-supplied market.	Accelerate tier-2 city restaurant acquisition targeting quality-first rather than volume-first strategy.
4	'Popular but Declining' restaurants (high volume, low rating) represent the highest platform risk -- their visibility brings customers but their poor ratings damage retention.	Trigger an immediate Quality Intervention Programme for all restaurants in this quadrant.
5	'Undiscovered Gems' (high rating, low volume) represent untapped growth -- they are not visible enough to attract the rating volume their quality deserves.	Create a 'New & Noteworthy' merchandising slot to boost these restaurants into higher discovery positions.
6	South Indian, Biryani, and Fast Food cuisine categories have the highest rating engagement (review volumes), indicating strongest consumer interest and repeat-order behaviour in these segments.	Increase marketing and promotional investment in these cuisine categories for maximum ROI.
7	Continental and Italian cuisine restaurants are disproportionately in the Premium tier but show no rating advantage over mid-range alternatives -- customers are not rewarding the price premium.	Advise premium Continental/Italian restaurants to improve experiential elements (packaging, speed) to justify pricing.
8	The rating distribution is compressed (most restaurants cluster 3.5-4.2), meaning that small algorithmic adjustments to how ratings are displayed can have outsized effects on restaurant discovery and order volumes.	Invest in rating display design improvements (e.g., showing rating trends, not just static scores) to differentiate restaurants more effectively.
9	IQR-based outlier analysis reveals that ultra-premium restaurants (> 3xIQR above Q3) attract negligible rating volumes, limiting their utility in training recommendation algorithms.	Exclude ultra-premium outliers from recommendation model training data to avoid skewing personalisation.
10	The dataset's top 10% of restaurants by rating_count collectively represent the majority of all ratings -- a Pareto concentration that creates algorithmic feedback loops, perpetually surfacing the same restaurants.	Introduce a diversity score in the recommendation engine to ensure newer, high-quality restaurants get fair discovery exposure.

11. Recommendations

Thefollowingfive recommendationsarespecific, actionable, and prioritised. Each maps directly from an insight to a recommended action with estimated business impact.

Priority	Insight	Recommendation	Expected Impact	Feasibility
1 (High)	Mid-range restaurants outperform in ratings and volume.	Launch a 'Swiggy Mid-Market Growth Programme' -- dedicated onboarding support and reduced commission for 6 months for qualifying mid-range restaurants in tier-2 cities.	+8-12% GMV in targeted cities within 12 months.	Medium -- requires sales team capacity and a commission waiver budget of ~INR 2-3 Cr.
2 (High)	'Popular but Declining' restaurants risk platform NPS.	Deploy a Quality Intervention Programme (QIP) -- automated alerts when rating drops >0.3 points over 30 days, followed by a dedicated account manager touchpoint and a 60-day improvement window before listing review.	-15% churn from at-risk restaurants; +0.05 platform average rating.	Medium -- requires CRM integration and trained account management staff.
3 (Medium)	Undiscovered Gems are under-surfaced.	Create a 'New & Noteworthy' home-page banner slot for restaurants with rating > 4.2 and rating_count < 50, rotated weekly. Track click-through and order conversion for 90 days.	+20% order volume for featured restaurants; improved platform rating diversity.	Low -- primarily a product/design change; no incremental cost.
4 (Medium)	Customers are value-conscious; price does not equal quality.	Replace the current 'Cost for Two' filter with a composite 'Value Score' (rating / price_tier_index) in the Swiggy discovery UI, helping customers find high-quality affordable options faster.	+5% session-to-order conversion for Mid-range segment.	Medium -- requires A/B test, product development sprint (~6-8 weeks).
5 (Low)	Cuisine engagement is uneven across cities.	Run cuisine-specific marketing campaigns (e.g., 'Biryani Fest', 'South Indian Week') in cities where that cuisine is under-represented but rated highly, to build supply-demand balance.	+10-15% orders in promoted cuisine categories during campaign period.	Low -- primarily a marketing spend and coordination effort.

12. Impact Estimation

The following impact estimates arebased on industry benchmarks for food delivery platforms, publicly available market data, and the analytical findings from this project. The logic is transparent and conservative; actual results will depend on execution quality.

Recommendation	Metric	Estimated Impact	Confidence
Mid-Market Growth Programme	Incremental GMV from new mid-range restaurants in tier-2 cities	INR 12-18 Cr annually (assuming 200 new restaurants x avg. INR 50K monthly GMV each)	Medium
Quality Intervention Programme	Reduction in restaurant churn + NPS improvement	INR 5-8 Cr retained GMV from at-risk restaurants; +2 NPS points platform-wide	Medium-High
Undiscovered Gems Visibility	Order volume uplift for featured restaurants	INR 1-2 Cr incremental GMV; near-zero implementation cost	High
Value Score Discovery Filter	Session-to-order conversion improvement	+5% conversion -- ~INR 8-12 Cr GMV at current platform volumes	Medium

WhyAct Now?

Swiggy is currently in a critical phase of post-IPO growth pressure, where improving unit economics and demonstrating platform quality are essential for investor confidence. Simultaneously, Zomato is aggressively expanding its Blinkit and dining-out offerings, increasing competitive pressure on food delivery. The recommendations above -- particularly the Quality Intervention Programme and Mid-Market Growth Programme -- address both the revenue opportunity and the competitive moat simultaneously, making the cost of inaction significantly higher than the cost of implementation.

13. Limitations

- **Staticdata:**The datasetisa point-in-time snapshot. Trends, seasonality, and time-series dynamics cannot be captured without longitudinal data.
- **No order-level data:** The analysis is based on listing attributes only. Actual GMV, conversion rates, repeat order frequency, and delivery performance cannot be measured.
- **Rating system opacity:** It is unknown how Swiggy's internal rating algorithm weights recency, photo uploads, or verified orders, which limits the interpretive confidence of raw ratings.
- **Self-selection bias:** Only restaurants that have successfully onboarded Swiggy are included; restaurants that were rejected or de-listed are absent, potentially skewing quality metrics upward.
- **Price field ambiguity:** 'Price for Two' is an estimated field, not a transaction-verified figure. Actual average order values may differ.
- **City classification:** Tier-1 / Tier-2 classification was applied based on domain knowledge, not an official government or industry taxonomy.
- **Causal inference limitations:** Correlations identified in this analysis (e.g., Mid-range restaurants have higher ratings) cannot be interpreted as causal relationships without controlled experimental data.

14. Future Scope

Additional Analysis With More Data

- Time-seriesanalysisofrestaurant ratingsover12-24 months to identify rating trajectory patterns (rising vs. declining).
- Customer cohort analysis linking order frequency, cuisine preference, and price sensitivity.
- Delivery time impact analysis -- correlating average delivery time with rating outcomes.
- Geospatial clustering of restaurant density vs. demand hotspots for new restaurant acquisition targeting.

New Data Sources That Would Strengthen Findings

- Swiggy order-leveldata (GMV,order frequency, cancellation rates)-- would enable direct revenue attribution.
- Google Maps / Zomato cross-platform ratings for the same restaurants -- would reveal platform-specific rating inflation.
- Census and mobile internet penetration data -- would refine tier-2 city opportunity sizing.
- Restaurant cost-of-goods and labour data -- would enable profitability modelling alongside rating quality.

Predictive Modelling & Real-Time Dashboard

Thecleaneddatasetis well-suitedfor supervised machine learning to predict restaurant rating trajectory. A gradient-boosted classification model (XGBoost / LightGBM) could predict whether a restaurant's rating will improve or decline over the next 90 days, enabling proactive merchant intervention. Additionally, a real-time Tableau or Power BI dashboard connected to a live database (PostgreSQL + REST API) could replace the current static analysis, giving the Growth team always-current restaurant performance visibility.

15. Conclusion

This project addressed a core business challenge for India's food delivery sector: understanding what drives customer satisfaction on Swiggy and where the platform's largest growth and quality improvement opportunities lie. Through a structured Python-based ETL pipeline, rigorous exploratory analysis, and statistical hypothesis testing, the team uncovered that price is a weak predictor of customer ratings -- Mid-range restaurants consistently outperform Budget and Premium segments -- and that significant geographic and cuisine-level performance gaps exist across the platform. The most impactful recommended action is the launch of a Mid-Market Growth Programme combined with a Quality Intervention Programme, which together are estimated to generate INR 17-26 Cr in incremental annual GMV while simultaneously improving the platform's average rating and customer NPS. All findings are reproducible via the GitHub-committed notebook pipeline and are visualised in an interactive Tableau dashboard designed for both executive and operational audiences.

16. Appendix

A. Data Dictionary (Full Column Definitions)

Column	DataType	Description	Valid Range / Values
restaurant_name	String	Official restaurant name as listed on Swiggy	Any string; no nulls allowed
city	String	City of restaurant operation	Any valid Indian city name
cuisine	String	Primary cuisine category	North Indian, South Indian, Chinese, Fast Food, Biryani, etc.
rating	Float	Customer satisfaction rating	0.0 - 5.0; rows with rating_count=0 excluded
rating_count	Integer	Total number of customer ratings	0 - unlimited; 0-count rows excluded
price_for_two	Integer	Estimated cost in INR for two people	Typically INR 50 - 3,000+
price_category	String (derived)	Price tier bucket	Budget / Mid-range / Premium
is_price_outlier	Boolean (derived)	IQR-based outlier flag for price field	True / False
rating_band	String (derived)	Rating quality bucket	Low (< 3.5) / Medium (3.5-4.0) / High (> 4.0)

B. Python ETL Logic Reference

```
#Column Standardisation
df.columns = [col.lower().replace(' ', '_') for col in df.columns]

# Remove nulls and zero-rating-count rows
df = df.dropna(subset=['rating', 'price_for_two', 'city', 'cuisine'])
df = df[df['rating_count'] > 0]

# Outlier detection (IQR method)
Q1 = df['price_for_two'].quantile(0.25)
Q3 = df['price_for_two'].quantile(0.75)
IQR = Q3 - Q1
df['is_price_outlier'] = (df['price_for_two'] < Q1 - 1.5*IQR) | (df['price_for_two'] > Q3 + 1.5*IQR)

# Price categorisation
def categorise_price(p):
    if p <= 200: return 'Budget'
    elif p <= 500: return 'Mid-range'
    else: return 'Premium'
df['price_category'] = df['price_for_two'].apply(categorise_price)

# Export
df.to_csv('data/processed/swiggy_cleaned_with_flags.csv', index=False)
df[~df['is_price_outlier']].to_csv('data/processed/swiggy_final_tableau.csv', index=False)
```

C. GitHub Repository Structure

```
Hopper_C-2_SwiggyDataAnalysis/

|-- data/raw/                    -- Original Swiggy dataset CSV
```

```
|-- data/processed/              -- Cleaned datasets (with flags + production-ready)
|-- notebooks/01_extraction.ipynb      -- Data loading and schema inspection
|-- notebooks/02_cleaning.ipynb        -- ETL and cleaning pipeline
|-- notebooks/03_eda.ipynb             -- Exploratory data analysis and visualisation
|-- notebooks/04_statistical_analysis.ipynb -- Hypothesis testing and segmentation
|-- notebooks/05_final_load_prep.ipynb -- Tableau export preparation
|-- scripts/etl_pipeline.py            -- Production ETL script
|-- reports/                     -- Static insight reports
|-- tableau/                     -- Tableau workbook files (.twbx)
|-- tableau/dashboard_links.md   -- Tableau Public URL
|-- docs/                        -- Additional documentation
|-- README.md
```

17. Contribution Matrix

Thismatrix documents eachteam member's contribution across all project phases. Primary = lead contributor; Secondary = supporting contributor. Claims are consistent with GitHub Insights, commit history, and PR records at github.com/kshitij2212/Hopper_C-2_SwiggyDataAnalysis.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau D ashboard	Report Writing	PPT & Viva
Kshitij Saxena			✓		✓		
Milan Kumar						✓	✓
Devanshu Prakash				✓	✓		
Rajat Tiwari	✓	✓					
Sahil Laxman Khemnar	✓				✓		

Note: GitHub contributors confirmed from repository: kshitij2212 (Kshitij Saxena - Team Lead) and milanchahar (Milan Kumar). Devanshu Prakash, Rajat Tiwari, and Sahil Laxman Khemnar contributed to offline analysis, report writing, and presentation preparation. Team members are requested to verify and sign off on this matrix at time of final submission.